

Ankush Kumar

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Skills

Technical skills : Python, Deep Learning, Pytorch, Nvidia NeMo, LangChain, Agentic AI, Megatron-LM, transformers (huggingface), slurm(basic), SQL, FastAPI.

Experience

Senior Technologist, Infosys Ltd (STG)

June 2018 – Present

Infosys Ltd - AI First Software Engineering(Generative AI)

- Currently working on Recursive Language Models (RLM) and reinforcement learning techniques to improve usability.
- Built the pipeline for Direct preference optimization(DPO) and by applying the DPO on in-house SLM achieved an improvement of 21% on benchmark over baseline model.
- Built a SOTA small language model for enterprise domains i.e., BankingSLM surpassing gemma-2-2b, llama-3.2-3B on finance benchmarks and powering finacle support pro application.
- Built a SOTA Small Language Model([SLM](#)) for code generation in Java i.e., NT-Java-1.1B which was accepted in NeurIPS AFM 2024.
- Developed a cutting-edge fine tuned code completion transformer based model for SAP ABAP using Parameter Efficient Fine-Tuning(PEFT) approach, employing Python, PyTorch, and DeepSpeed to optimize training time and performance.

Infosys Ltd - Document Digitization

- Constructed a tagging tool to streamline the annotation process of digitized documents, significantly reducing time consumption for processing a large volume of documents by 95% per document.
- Led a team of three individuals in generating a new dataset to demonstrate the proof of concept (POC) for an open-sourced machine learning model (LayoutLMV2).
- Successfully fine-tuned the model using a specific use-case to optimize its performance by ~60 accuracy.

Retail Clothing Brand - Modernization(Data and Reporting)

- Designed and architected a robust data transformation framework specifically tailored for the reporting track, enabling the migration from MicroStrategy to PowerBI. This resulted in a remarkable 96% reduction in report availability time for multiple reports.
- Developed efficient pipelines to automate the extract-transform-load (ETL) process for analytics and reports by leveraging the data build tool (dbt). Utilized Azure Synapse as the data analytics service, ensuring streamlined data transformation workflows.

American Multinational Financial Services Corporation - Anomaly Detection

- Implemented a comprehensive end-to-end machine learning solution, utilizing clustering techniques, for detecting anomalies in offers. Leveraged Python and PySpark on ML Studio to identify and isolate anomalous offers caused by manual or system errors.

Education, Certification & Publications

Bachelor of Technology, Computer Science and Engineering
[Narrow Transformer: Mono-lingual Code SLM for Desktop](#)(NeuIPS AFM 2024)
[State-of-the-art Small Language Coder Model: Mify-Coder](#)
[Architecture for Retail Report Data Generation On Time Periods](#)(Medium)

Awards & Recognition

- Receive Stock Award for the last 3 consecutive years.
- Received best mentor award for power interns in Infosys.
- Received the prestigious INSTA Award for Narrow Transformer solutions.
- Received the prestigious INSTA Award for exceptional performance in implementing and delivering end-to-end anomaly detection solutions.
- Honored with the INSTA Award for exemplary performance in unit leadership, specifically for architecting the ETL process for Retail.